

Research Notes on Decision tree

Decision tree

>> Section 1

Decision trees can also be seen as generative models of induction rules from empirical data. An optimal decision tree is then defined as a tree that accounts for most of the data, while minimizing the number of levels (or "questions"). Several algorithms to generate such optimal trees have been devised, such as ID3/4/5, CLS, ASSISTANT, and CART.

>> Section 2

Now, we can use the formulas to calculate the phi function values and information gain values for each M in the dataset. Once all the values are calculated the tree can be produced. The first thing to be done is to select the root node. In information gain and the phi function we consider the optimal split to be the mutation that produces the highest value for information gain or the phi function. Now assume that M1 has the highest phi function value and M4 has the highest information gain value. The M1 mutation will be the root of our phi function tree and M4 will be the root of our information gain tree. You can observe the root nodes below

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They are unstable, meaning that a small change in the data can lead to a large change in the structure of the optimal decision tree. They are often relatively inaccurate. Many other predictors perform better with similar data. This can be remedied by replacing a single decision tree with a random forest of decision trees, but a random forest is not as easy to interpret as a single decision tree. For data including categorical variables with different numbers of levels, information gain in decision trees is biased in favor of those attributes with more levels. Calculations can get very complex, particularly if many values are uncertain and/or if many outcomes are linked.

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The tree using information gain has the same results when using the phi function when calculating the accuracy. When we classify the samples based on the model using information gain we get one true positive, one false positive, zero false negatives, and four true negatives. For the model using the phi function we get two true positives, zero false positives, one false negative, and three true negatives. The next step is to evaluate the effectiveness of the decision tree using some key metrics that will be discussed in the evaluating a decision tree section below. The metrics that will be discussed below can help determine the next steps to be taken when optimizing the decision tree.

>> Section 5

A decision tree is a decision support recursive partitioning structure that uses a tree-like model of decisions and their possible consequences, including chance event outcomes, resource costs, and utility. It is one way to display an algorithm that only contains conditional control statements. Decision trees are commonly used in operations research, specifically in decision analysis, to help identify a strategy most likely to reach a goal, but are also a popular tool in machine learning.

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if condition1 and condition2 and condition3 then outcome. Decision rules can be generated by constructing association rules with the target variable on the right. They can also denote temporal or causal relations.

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Drawn from left to right, a decision tree has only burst nodes (splitting paths) but no sink nodes (converging paths). So used manually they can grow very big and are then often hard to draw fully by hand. Traditionally, decision trees have been created manually – as the aside example shows – although increasingly, specialized software is employed.

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Other techniques The above information is not where it ends for building and optimizing a decision tree. There are many techniques for improving the decision tree classification models we build. One of the techniques is making our decision tree model from a bootstrapped dataset. The bootstrapped dataset helps remove the bias that occurs when building a decision tree model with the same data the model is tested with. The ability to leverage the power of random forests can also help significantly improve the overall accuracy of the model being built. This method generates many decisions from many decision trees and tallies up the votes from each decision tree to make the final classification. There are many techniques, but the main objective is to test building your decision tree model in different ways to make sure it reaches the highest performance level possible.

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Overview A decision tree is a flowchart-like structure in which each internal node represents a test on an attribute (e.g. whether a coin flip comes up heads or tails), each branch represents the outcome of the test, and each leaf node represents a class label (decision taken after computing all attributes). The paths from root to leaf represent classification rules. In decision analysis, a decision tree and the closely related influence diagram are

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used as a visual and analytical decision support tool, where the expected values (or expected utility) of competing alternatives are calculated. A decision tree consists of three types of nodes:

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We will set D , which is the depth of the decision tree we are building, to three ($D = 3$). We also have the following data set of cancer and non-cancer samples and the mutation features that the samples either have or do not have. If a sample has a feature mutation then the sample is positive for that mutation, and it will be represented by one. If a sample does not have a feature mutation then the sample is negative for that mutation, and it will be represented by zero. To summarize, C stands for cancer and NC stands for non-cancer. The letter M stands for mutation, and if a sample has a particular mutation it will show up in the table as a one and otherwise zero.

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This is the phi function formula. The phi function is maximized when the chosen feature splits the samples in a way that produces homogenous splits and have around the same number of samples in each split.

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Once we have calculated the key metrics we can make some initial conclusions on the performance of the decision tree model built. The accuracy that we calculated was 71.60%. The accuracy value is good to start but we would like to get our models as accurate as possible while maintaining the overall performance. The sensitivity value of 19.64% means that out of everyone who was actually positive for cancer tested positive. If we look at the specificity value of 99.06% we know that out of all the samples that were negative for cancer actually tested negative. When it comes to sensitivity and specificity it is important to have a balance between the two values, so if we can decrease our specificity to increase the sensitivity that would prove to be beneficial. These are just a few examples on how to use these values and the meanings behind them to evaluate the decision tree model and improve upon the next iteration.